

**BEFORE YOU READ — ANSWER FROM MEMORY**

1. What two things can switch pepsinogen on, and which enzyme activates the other pancreatic proteases?
2. Which GI hormone was the first to be discovered, and what releases it?
3. Why does lactose left in the intestine cause both diarrhoea and cramps?

Guessing wrong before reading still improves retention — attempt all three, then check as you go.

1 Digestion in one glance

Digestion	Breakdown of large insoluble molecules into smaller soluble molecules that pass through the gut wall into the blood
Two kinds	Mechanical — food broken down by movement · Chemical — food broken down by a chemical or enzyme
Accessory organs	Teeth, tongue, liver, pancreas, gall bladder, salivary glands — food does not pass through them ; they assist digestion but are not part of the alimentary canal

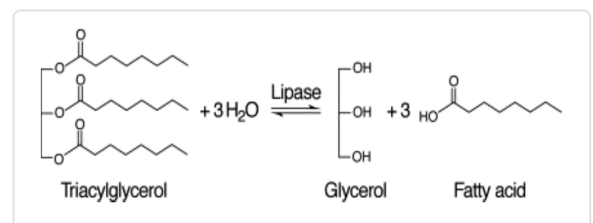
2 Enzymes — general properties

What they are	Biocatalysts that speed up biochemical reactions without being used up
Structure	Water-soluble , highly specialised proteins; most are globular proteins
How they work	Lower the activation energy of the reaction; they do not affect the nature or properties of the end product
Specificity	Highly specific — each enzyme catalyses one kind of substrate
Ribozymes	RNA with catalytic activity (name = RNA + enzyme); act by transesterification and phosphodiester bond hydrolysis; inactivated by heating above their melting temperature or if essential nucleotides are changed

3 Digestive enzymes — the families

Job	Split the large, complex molecules of proteins, carbs and fats into smaller ones that can be absorbed into the blood
Secreted by	Salivary glands and cells lining the stomach, pancreas and small intestine
The six named	Amylase · maltase · lactase · sucrase · lipase · proteases — some foods need a particular enzyme to break down the nutrients they contain

Family	Reaction	Where it comes from
Amylases	Starch → dextrin* , maltose and glucose	Salivary gland, pancreas
Lipases	Fats → fatty acids + glycerol	Small amounts from mouth and stomach; larger amounts from the pancreas
Proteases	Proteins → smaller polypeptides or single amino acids	Stomach and pancreas ; also called peptidases, proteolytic enzymes, proteinases



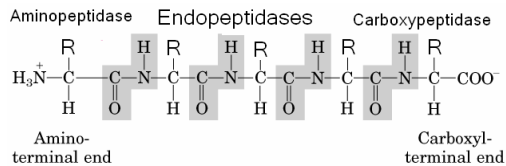
Lecture figure — lipase: triacylglycerol + 3 H₂O ⇌ glycerol + 3 fatty acids.

* The slide prints this product as “dextran”.

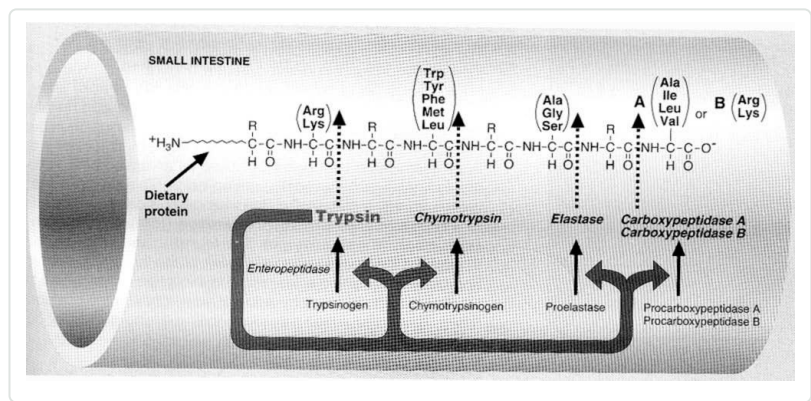
4 Proteases — who cuts where

Two types of GIT proteolytic enzyme

Endopeptidases	Hydrolyse the inner peptide bonds
Exopeptidases: aminopeptidases	Bond of the N-terminal amino acid
Exopeptidases: carboxypeptidases	Bond of the C-terminal amino acid



Lecture figure — aminopeptidase at the amino end, endopeptidases inside, carboxypeptidase at the carboxyl end.



Lecture figure — enteropeptidase turns trypsinogen into trypsin; trypsin then activates chymotrypsinogen, proelastase and procarboxypeptidases A and B. Dotted arrows show where each enzyme cuts.

Enzyme	Source and activation	Cuts at	Remember
Pepsin	Stomach — secreted as the inactive zymogen pepsinogen ; activated by HCl or autocatalytically by pepsin already active	Acid-stable endopeptidase	Releases peptides and a few free amino acids from dietary protein
Rennin	—	Milk protein (casein)	In infants
Trypsin	Secreted by the pancreas ; works in the 1st section of the small intestine	Only where the carbonyl group comes from Arg or Lys	Starts protein digestion there, then activates carboxypeptidase and chymotrypsin
Chymotrypsin	From chymotrypsinogen (figure)	Trp, Tyr, Phe, Met, Leu	Breaks peptides into free amino acids that the intestinal wall can absorb
Elastase	From proelastase (figure)	Ala, Gly, Ser	—
Carboxypeptidase A	Pancreas	Ala, Ile, Leu, Val	Splits peptides into individual amino acids
Carboxypeptidase B	Pancreas	Arg and Lys (basic amino acids)	Same residues as trypsin — but B works at the C-terminal end

5 When enzymes are missing

Causes of deficiency	Diet high in sugar and processed foods · repeated antibiotic use · smoking · aging · pregnancy · physical, psychological and emotional stress. Some deficiencies are inherited ; others develop over time
Eight signs	Food intolerances · abdominal pain or discomfort · diarrhoea and bloating · constipation and undigested food in the stools · poor nutrient absorption · weaker immune system · fatigue, anxiety and depression · skin problems

LACTOSE LEFT IN THE INTESTINE

Diarrhoea	↑ osmotic pressure → water drawn from the tissue (dehydration) into the large intestine
Cramps	Intestinal bacteria ferment lactose → CO ₂ gas → abdominal cramps
Treatment	Remove lactose (milk) from the diet

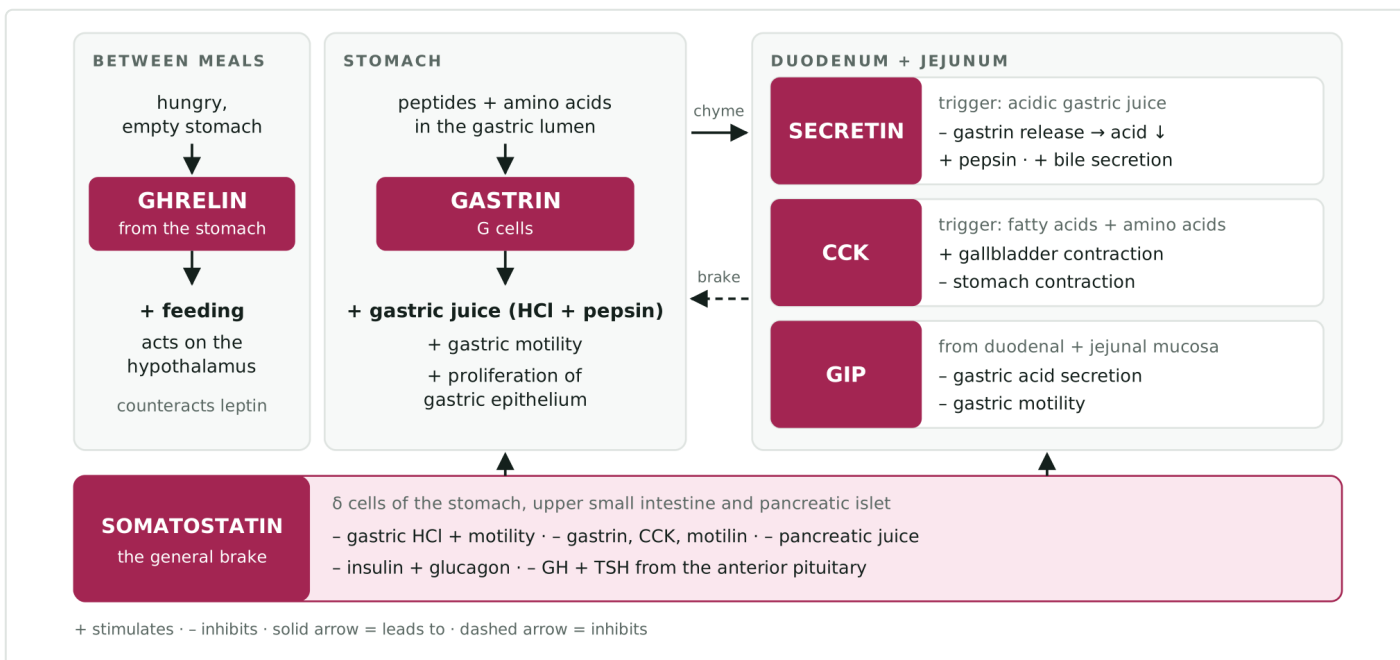
EXOCRINE PANCREATIC INSUFFICIENCY (EPI)

Fault	Deficiency of the pancreas's key enzymes — amylase, protease, lipase
Result	Food is not digested properly, especially fats
Seen with	Chronic pancreatitis (can permanently damage the organ) · cystic fibrosis (inherited; damages lungs and digestive system, including the pancreas) · pancreatic cancer

6 Hormones — general properties

What a hormone is	A chemical substance released by a cell or gland in one part of the body that sends messages to cells in other parts — a chemical messenger carrying a signal from one cell to another
Shared features	Secreted in very small amounts · transported, usually, in the blood · target cells have specific receptors · regulate cell reactions by affecting gene expression (often gene transcription factors) · regulate metabolic events

7 The six GIT hormones — one meal, start to finish



Concept map built from the lecture's hormone slides: gastrin drives the stomach; the duodenal hormones and somatostatin hold it back.

Hormone	Made by	Released by	Actions	ID fact
1 Gastrin	G cells of the stomach, duodenum and pancreas	A meal — peptides and amino acids in the gastric lumen	↑ gastric juice (HCl + pepsin); ↑ gastric motility; proliferation of gastric epithelium	Peptide hormone
2 Cholecysto-kinin (CCK)	Duodenum and jejunum	Digestive products of fat, fatty acids, protein and monoglycerides	Strongly contracts the gallbladder → bile into the small intestine, where it emulsifies fat; inhibits stomach contraction	Peptide hormone
3 Secretin	Duodenum and upper jejunum	Acidic gastric juice emptying into the duodenum	↓ gastric acid by inhibiting gastrin release from G cells; ↑ pepsin secretion; ↑ bile secretion; controls duodenal pH (acid control + bicarbonate buffering); regulates water homeostasis	First GI hormone discovered
4 Somatostatin	δ cells of the stomach, upper small intestine and pancreatic islet	—	Inhibits GH and TSH (anterior pituitary), gastric HCl secretion and motility, pancreatic juice, insulin and glucagon , and GI hormones (gastrin, CCK, motilin)	"GH release inhibitory factor"; first found in the hypothalamus ; two forms — 14 and 28 amino acids
5 GIP	Duodenum and jejunum mucosa	—	Inhibits gastric acid secretion and gastric motility	Gastric inhibitory peptide
6 Ghrelin	Endocrine cells in the stomach	Hunger	Acts on the hypothalamus to stimulate feeding; counteracts the inhibition of feeding by leptin	Lipopeptide — 28 amino acids + a covalently attached 8-carbon fatty acid

Discriminators — the pairs that get confused

Gastric acid	Only gastrin raises it. Secretin (through gastrin), somatostatin and GIP all lower it
Stomach movement	Gastrin speeds it up; CCK inhibits stomach contraction; somatostatin and GIP lower gastric motility
Bile	CCK contracts the gallbladder; secretin stimulates the secretion of bile
"28 amino acids"	Somatostatin (one of its two forms) or ghrelin — only ghrelin carries the 8-carbon fatty acid
Hypothalamus	Somatostatin was first found there; ghrelin acts there

P Must-remember pearls

- **Pepsinogen → pepsin** by HCl or by pepsin already active. Trypsin is the pancreatic switch — it activates carboxypeptidase and chymotrypsin.
- **Arg/Lys appears twice:** trypsin cuts there inside the chain; carboxypeptidase B removes them from the C-terminal end. Chymotrypsin = Trp, Tyr, Phe, Met, Leu. Elastase = Ala, Gly, Ser. Carboxypeptidase A = Ala, Ile, Leu, Val.
- **Lactose left in the gut:** osmotic pressure → diarrhoea and dehydration; bacterial fermentation → CO₂ → cramps. Treatment is removing milk.
- **EPI** = too little amylase, protease and lipase; fats suffer most. Think chronic pancreatitis, cystic fibrosis, pancreatic cancer.
- **Gastrin is the accelerator** (HCl + pepsin, motility). Secretin — first GI hormone discovered, released by acid — somatostatin and GIP are the brakes on acid. CCK empties the gallbladder.

Q Test yourself

1. A man gets painful abdominal cramps after every glass of milk. Which process produces the cramps?

- A) Water drawn into the large intestine
B) Bacterial fermentation of lactose to CO₂
C) Loss of pancreatic lipase
D) Excess gastric acid

2. A child with cystic fibrosis cannot digest food properly, especially fats. Which enzymes are deficient?

- A) Pepsin and rennin
B) Maltase, lactase and sucrase
C) Amylase, protease and lipase
D) Aminopeptidases only

3. A pancreatic enzyme removes arginine from the C-terminal end of a peptide. Which enzyme is it?

- A) Trypsin
B) Carboxypeptidase A
C) Elastase
D) Carboxypeptidase B

4. GIP is secreted by the mucosa of the duodenum and jejunum. What does it do?

- A) Inhibits gastric acid secretion and gastric motility
B) Contracts the gallbladder
C) Stimulates feeding
D) Stimulates pepsin secretion

5. In one patient a single hormone is suppressing both insulin and glucagon secretion. Which hormone?

- A) Gastrin
B) Secretin
C) Somatostatin
D) Ghrelin

ANSWERS — COVER UNTIL YOU'VE COMMITTED

- 1. B — Bacterial fermentation of lactose to CO₂** — gas causes the cramps; the osmotic pull of water causes the diarrhoea.
2. C — Amylase, protease and lipase — exocrine pancreatic insufficiency, which cystic fibrosis can cause.
3. D — Carboxypeptidase B — B takes Arg and Lys off the C-terminal end; trypsin cuts at them inside the chain.
4. A — Inhibits gastric acid secretion and gastric motility — GIP is gastric inhibitory peptide.
5. C — Somatostatin — it also inhibits GH, TSH, gastric acid and several GI hormones.

SPACED REVIEW

☐ Day 1

☐ Day 3

☐ Day 7

☐ Day 21

☐ Pre-exam